

Tunneling in Thermal Unimolecular Reactions. Formaldehyde

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Using available data, mostly photochemical, at discrete energies near threshold for reaction, as well as thermochemical data and results of ab initio calculations, I have computed specific-energy rate constants for the decomposition of formaldehyde into $\text{H}_2 + \text{CO}$, which involves tunneling, and into $\text{H} + \text{HCO}$, which does not. The corresponding thermal rate constants and activation energies are then obtained and it is shown that tunneling causes the thermal rate and activation energy to fall off "forever" with pressure. Despite the uncertainties regarding some of the experimental data and theoretical parameters, it is concluded that the photochemical and thermal experimental data are reasonably consistent with one another, in particular with the notion that the decomposition into $\text{H}_2 + \text{CO}$ proceeds by tunneling.

1. Introduction

Tunneling, i.e., the nonclassical passage through a potential energy barrier, is well studied and well documented in bimolecular reactions, particularly hydrogen-transfer reactions,¹ as is the associated kinetic isotope effect.² By contrast, tunneling in unimolecular reactions has received only very little attention until recently, when it was invoked to account for energy partitioning in hydrogen-atom elimination from³ $\text{CH}_2\text{CH}_2\text{F}$, and in connection with the decomposition of formaldehyde into^{4,5} $\text{H}_2 + \text{CO}$.

The purpose of this work is to investigate the effect of quantum-mechanical tunneling in thermal unimolecular reactions, notably as it manifests itself in the pressure dependence of the (thermal) rate constant and activation energy, and with particular attention to the specific case of formaldehyde. The experimental information available at present on the formaldehyde system is twofold: laser photochemical data at discrete energies near the energy threshold for reaction and high-temperature (shock-tube) thermal data. The present work aims to provide a link between the two sets of data and should reveal to what extent the laser data are compatible with thermal data, and vice versa. This is of some practical interest since formaldehyde is an important intermediate in the oxidation of hydrocarbons.⁶ A preliminary account has appeared.^{7,8}

2. Theory

The standard (RRKM) expression for the unimolecular rate constant $k(E)$ for decomposition of a molecule having specified energy E relative to its ground state can be written quite generally as

$$k(E) = \alpha G^*(E) / [hN(E)] \text{ s}^{-1} \quad (1)$$

Here $N(E)$ is the density of states (number of states per unit energy) of the molecule at E , α is the reaction path

degeneracy, and $G^*(E)$ is the number of ways that available energy in the transition state (asterisk in superscript shall henceforth denote properties of the transition state) can be shared between internal states of the transition state and relative translational energy of fragments.

If E_0 is the height of the energy barrier separating reactant from fragments, i.e., the difference between ground-state energies of the transition state and the reactant, and if no tunneling through the barrier occurs, relative fragment translation cannot take place unless $E > E_0$; then semiclassically

$$G^*(E) = \int_{E_0}^E N^*(E-x) dx = G^*(E-E_0) \quad E > E_0 \quad (2)$$

where $N^*(E-x)$ is the density of states of the transition state at energy $E-x$. (If $E < E_0$, $G^*(E) = 0$ and also $k(E) = 0$.) Equation 2 is the usual RRKM result. If tunneling through the barrier does occur, relative translation of fragments can take place even if $E < E_0$, although with reduced probability. Equation 2 then becomes

$$G^*(E) = \int_0^E p(x) N^*(E-x) dx \quad E > 0 \quad (3)$$

where $p(x)$ is the probability (commonly referred to as tunneling probability) that fragments shall appear with relative translational energy x . Note that $G^*(E)$ is now finite everywhere (although perhaps very small). Equation 3 has been derived previously^{3,4} in a slightly different form; it represents the simplest possible way of accounting for tunneling and amounts merely to a generalization of eq 2. Equation 2 can in fact be immediately recognized as a special form of eq 3 for the case when $p(x)$ is the step function: $p(x) = 0$ if $x < E_0$, and $p(x) = 1$ if $x > E_0$, which corresponds to absence of tunneling.

The rate constant including tunneling follows by substituting eq 3 into eq 1. Since $G^*(E)$ is then finite everywhere, so is $k(E)$, in contrast with the more usual case of no tunneling where $k(E)$ is zero for all energies below threshold E_0 . In other words, tunneling causes the sharp energy threshold for reaction to disappear (Figure 1), with interesting consequences for the pressure dependence of the thermal rate constant (see section 3).

A connection between the specific-energy rate constant $k(E)$ and thermal properties is provided by the thermal unimolecular limiting high-pressure rate constant k_∞ which is the Boltzmann average of $k(E)$:

$$k_\infty = \frac{1}{Q} \int_0^\infty k(E) N(E) e^{-E/kT} dE = \frac{1}{Q} \mathcal{L}\{k(E) N(E)\} \quad (4)$$

(1) R. P. Bell, "The Tunnel Effect in Chemistry", Chapman and Hall, London, 1980.

(2) See, for example, "Isotope Effect in Chemical Reactions", C. J. Collins and N. S. Bowman, Eds., ACS Monograph 167, Van Nostrand-Reinhold, New York, 1970.

(3) S. Kato and K. Morokuma, *J. Chem. Phys.*, **72**, 206 (1980).

(4) W. H. Miller, *J. Am. Chem. Soc.*, **101**, 6810 (1979).

(5) S. K. Gray, W. H. Miller, Y. Yamaguchi, and H. F. Schaefer, *J. Am. Chem. Soc.*, **103**, 1900 (1981).

(6) See, for example, S. W. Benson, "The Foundation of Chemical Kinetics", McGraw-Hill, New York, 1960, p 479 ff.

(7) W. Forst, *Oxidation Commun.*, in press.

(8) W. Forst, paper presented at the 7th International Symposium on Gas Kinetics, Göttingen, Aug 1982.

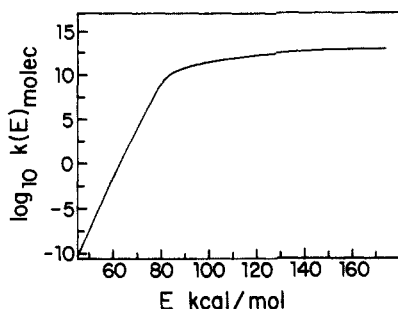


Figure 1. Specific-energy rate constant $k(E)$ for formaldehyde molecular channel, calculated from eq 1, 3, and 21 for $E_0 = 81$ kcal/mol, $D = 94.38$ and molecular and transition-state frequencies from Table I.

where $\mathcal{L}\{\dots\}$ represents the Laplace transform. The integral in eq 3 is the so-called convolution integral, commonly represented symbolically by $p(E) \cdot N^*(E)$. Using this notation, and substituting eq 3 and 1 into eq 4, we have

$$k_\infty = (1/hQ) \mathcal{L}\{p(E) \cdot N^*(E)\} \quad (5)$$

It can be shown⁹ that the partition function Q of the reactant molecule is the Laplace transform of $N(E)$; similarly

$$\mathcal{L}\{N^*(E)\} = Q^*(s) \quad (6)$$

where $Q^*(s)$ is the partition function of the transition state, written to show explicitly its dependence on $s = 1/kT$, the transform parameter. We shall write similarly $Q(s)$ for Q . Now if

$$\mathcal{L}\{p(E)\} = P(s) \quad (7)$$

then it can be shown from standard Laplace transform theory¹⁰ that, given eq 6 and 7, it follows that

$$\mathcal{L}\{p(E) \cdot N^*(E)\} = P(s) Q^*(s) \quad (8)$$

and hence, from eq 5 and 8

$$k_\infty = \frac{P(s)}{h} \frac{Q^*(s)}{Q(s)} \quad (9)$$

which is the most general transition-state expression for k_∞ . If tunneling does not occur, $p(E)$ is the step function previously described, the Laplace transform of which is¹¹

$$P(s) = e^{-sE_0}/s = kT e^{-E_0/kT} \quad (10)$$

and hence

$$k_\infty = \frac{kT}{h} \frac{Q^*(s)}{Q(s)} e^{-E_0/kT} \quad (11)$$

the familiar result of transition-state theory without tunneling.⁹ In the literature,¹² $\Gamma^* = sP(s)e^{E_0/kT}$ is referred to as the (thermal) tunneling correction factor.

In many cases (including formaldehyde discussed later), the energy barrier separating reactant from fragments can be approximated by an inverted parabola,¹ for which, using the same energy zero as in eq 1-3

$$p(E) = \frac{1}{2} \left\{ 1 + \tanh \frac{D}{2} \left(\frac{E}{E_0} - 1 \right) \right\} \quad (12)$$

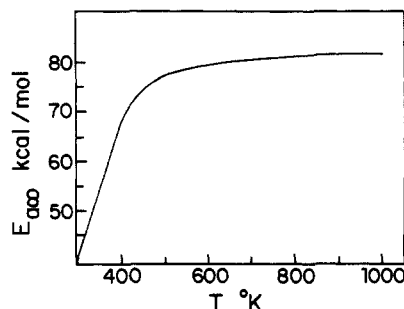


Figure 2. Limiting high-pressure activation energy E_∞ for formaldehyde molecular channel, calculated from eq 7, 16, and 21. Same parameters as in Figure 1. In this case $\beta/s = 1$ corresponds to $T = 432$ K.

where D is a parameter related to the thickness of the barrier. Equation 7 then yields, after some manipulation

$$P(s) = \frac{e^{-sE_0}}{s} \int_0^{\exp(sE_0)} \frac{dx}{1 + x^{\beta/s}} \quad (13)$$

where $\beta/s = DkT/E_0$. This integral is solvable analytically¹³ for certain values of β/s , yielding fairly complicated expressions which can be approximated reasonably well by^{1,14}

$$P(s) \sim \frac{E_0 e^{-D}}{sE_0 - D} \quad \beta/s < 1 \quad (14)$$

$$\sim \frac{\pi e^{-sE_0}}{\beta \sin \frac{\pi s}{\beta}} \quad \beta/s > 1 \quad (15)$$

The activation energy is, from eq 9

$$E_\infty = -\frac{d \ln k_\infty}{ds} = -\frac{dP(s)}{ds} \frac{1}{P(s)} + \langle E^* \rangle - \langle E \rangle \quad (16)$$

where $\langle E^* \rangle$ and $\langle E \rangle$ are the average energies of the transition state and the reactant molecule, respectively. From eq 14 and 15 we have approximately

$$-\frac{dP(s)}{ds} \frac{1}{P(s)} \sim \frac{E_0}{sE_0 - D} \quad \beta/s < 1 \quad (17)$$

$$\sim E_0 + \frac{\pi E_0}{D} \cot \left(\frac{\pi s E_0}{Dk} \right) \quad \beta/s > 1 \quad (18)$$

which shows that for $\beta/s < 1$, i.e., for $T < E_0/Dk$, the logarithmic derivative of $P(s)$ increases roughly linearly with T , whereas for $\beta/s > 1$ it is virtually constant at $\sim E_0$. Since the temperature dependence of $\langle E^* \rangle - \langle E \rangle$ is only modest, we see that tunneling produces a strong temperature dependence of E_∞ below $T \sim E_0/Dk$ (Figure 2). The result is of course an appreciable curvature in the Arrhenius plot of k_∞ at low temperatures. All of these effects of tunneling are analogous to those found in bimolecular reactions.

The pressure-dependent unimolecular rate constant $k_{\text{uni}}(s^{-1})$

$$k_{\text{uni}} = \frac{\omega}{Q} \int_0^\infty \frac{k(E)}{k(E) + \omega} N(E) e^{-E/kT} dE \quad (19)$$

(9) W. Forst, "Theory of Unimolecular Reactions", Academic Press, New York, 1973.

(10) D. V. Widder, "The Laplace Transform", Princeton University Press, Princeton, NJ, 1971.

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(12) H. S. Johnston and D. Rapp, *J. Am. Chem. Soc.*, **83**, 1 (1961).

(13) I. S. Gradshteyn and I. M. Ryzhik, "Table of Integrals, Series and Products", 4th ed. Translated by A. Jeffrey, Academic Press, New York, 1965, p 63 ff.

(14) E. E. Nikitin, "Theory of Elementary Atomic and Molecular Processes in Gases", Translated by M. J. Kearsley, Oxford University Press, 1974, p 22; also ref 1, Appendix C.

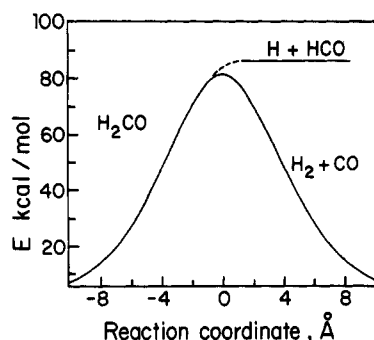


Figure 3. Eckart barrier for formaldehyde molecular channel calculated from eq 22. Same parameters as in Figure 1, with μ = reduced mass of H_2/CO system. Shown also is the radical-channel barrier at 86 kcal/mol, which is meant to be only schematic.

where ω = collision frequency (this is the so-called strong-collision version of k_{uni}) is not amenable to analytical solution and the effect of tunneling can only be determined numerically in specific cases. The pressure-dependent activation energy E_a is the temperature coefficient of k_{uni} , defined in a manner similar to eq 16, i.e.

$$E_a = kT^2(d \ln k_{\text{uni}}/dT) \quad (20)$$

3. Decomposition of Formaldehyde into $\text{CO} + \text{H}_2$

Although the reaction $\text{H}_2\text{CO} \rightarrow \text{CO} + \text{H}_2$ (to be referred to as the molecular channel) is very nearly thermoneutral,¹⁵ there is a large energy barrier separating the reactant and products, for which a recent ab initio calculation⁵ gives the (zero-point corrected) value $E_0 = 87$ kcal/mol. The laser photochemistry of formaldehyde has received a great deal of attention and appears to be quite complex,¹⁶ but it seems reasonably certain that the molecular products $\text{H}_2 + \text{CO}$ begin to form the ground-state formaldehyde at laser wavelengths of ~ 355 nm = 80 kcal/mol, i.e., at energies below the top of the calculated energy barrier, thus indicating tunneling.

The fact that the molecular-channel decomposition is thermoneutral suggests a symmetrical energy barrier. If we use the one-dimensional symmetrical Eckart barrier, we have¹²

$$p(E) = \frac{\cosh A - 1}{\cosh A + \cosh B} \quad (21)$$

where $A = 2D(E/E_0)^{1/2}$, $B = (4D^2 - \pi^2)^{1/2}$, $D = 2\pi E_0/|\hbar\nu^*|$, $\hbar\nu^*$ being the imaginary frequency of the transition-state oscillator undergoing dissociation. Miller, Schaefer, et al.⁵ calculate $\hbar\nu^* = 2026i$, and this combined with their value $E_0 = 87$ kcal/mol yields $D = 94.38$, which is quite high. Since the Eckart barrier itself is given by¹

$$V(x) = E_0 \text{sech}^2 [(2x/Dh)(2\mu E_0)^{1/2}] \quad (22)$$

where x is a distance along the reaction coordinate and μ is the reduced mass of the tunneling particles, we see that large D and E_0 means a high but thin barrier (Figure 3)—only about 4 Å thick at ~ 10 kcal/mol below the top of the barrier. One consequence of this type of barrier is that it can be fairly well approximated by a parabolic barrier so that eq 12 and 21 yield very similar results except at very low energies (for $k(E)$) or low temperatures (for $P(s)$).

For the purpose of evaluating $G^*(E)$, the method of steepest descents was used to generate $N^*(E)$, using

TABLE I: Vibrational Frequencies^a in cm^{-1}

formaldehyde molecule ^b	transition states	
	molecular channel ^c	radical channel ^d
2944	3125	2200 (2)
1764	1830	1000
1563	1523	plus two one-dimensional
1191	839	rotors $B = 1.2 \text{ cm}^{-1}$
3009	936	
1287		

^a Degeneracies in parentheses if different from 1.

^b Reference 39. ^c Reference 5. ^d This work.

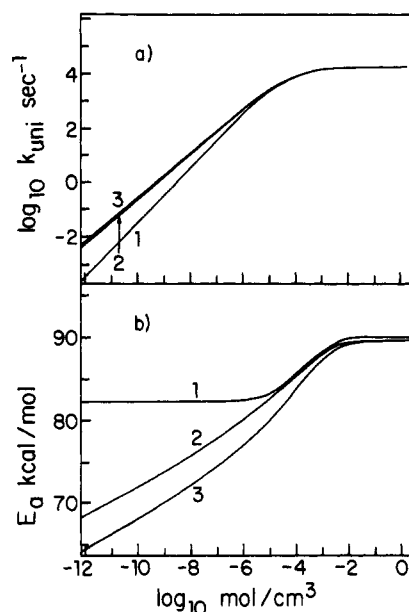


Figure 4. Falloff at 2000 K of thermal unimolecular rate constant k_{uni} (panel a) and activation energy E_a (panel b) for formaldehyde molecular-channel decomposition only, calculated from eq 19 and 20, respectively, using $E_0 = 87$ kcal/mol; other parameters the same as in Figure 1, except for the following: curve 1, $D \rightarrow \infty$ (no tunneling); curve 2, $D = 100$; curve 3, $D = 94.38$.

transition-state frequencies given in Table I, and $p(E)$ was taken to be that of eq 21, with the cited values of D and E_0 ; the integral in eq 3 was then evaluated by 64-point Gaussian quadrature.¹⁷ Use of the Bayer-Swinehart algorithm¹⁸ for generating exact $N^*(E)$ did not produce significantly different $G^*(E)$, so that the procedure was not deemed worth the added machine time and all the calculations reported here are based on the steepest-descent method. This goes also for $N(E)$ in eq 1, for which the molecular frequencies are also listed in Table I. The specific-energy rate constant $k(E)$ obtained in this way from eq 1, 3, and 21 is shown in Figure 1.

The thermal rate constant k_{uni} and activation energy E_a then follow from eq 19 and 20. The integrations involved were done by 128-point Gauss-Laguerre quadrature.¹⁹ The collision frequency ω was calculated for hard-sphere collisions between formaldehyde and a particle of mass 40 (argon) using a mean collision diameter of 3.6 Å. The results of calculations are shown in Figure 4 for three different values of D , one of which is $D \rightarrow \infty$ which corresponds to no tunneling.

(17) M. Abramowitz and I. A. Stegun, "Handbook of Mathematical Functions", NBS Applied Mathematics Series 55.

(18) S. E. Stein and B. S. Rabinovitch, *J. Chem. Phys.*, **58**, 2438 (1973).

(19) G. Emanuel, "Table of the Kassel Integral", Aerospace Report No. TR-0200 (4240-20)-5, 1969, Appendix, NTIS, Springfield, VA, No. AD 685-157.

(15) S. W. Benson, "Thermochemical Kinetics", 2nd ed., Wiley, New York, 1976.

(16) For a recent summary see P. Ho, D. J. Bamford, R. J. Buss, Y. T. Lee, and C. B. Moore, *J. Chem. Phys.*, **76**, 3630 (1982).

One is immediately struck by the fact that, while the activation energy falloff makes it perfectly clear that below $\sim 10^{-6}$ mol/cm³ the reaction in the absence of tunneling is at its low-pressure limit and that when tunneling is present falloff continues even at the lowest pressures, none of this is obvious from the rate constant falloff. Since, in the last analysis, the pressure dependence of E_a and of k_{uni} is merely the reflection of the energy dependence of $k(E)$, these results show that the activation energy falloff is a sensitive indicator of the energy dependence of $k(E)$, while the falloff of k_{uni} is not. This point has been made before and merits additional emphasis.²⁰

The second important conclusion to be drawn from Figure 4 is that, when tunneling operates, both E_a and k_{uni} fall off "forever". This is because the low-pressure limit of a unimolecular reaction is defined by the condition $\omega \ll k(E)$ for every E . Since in the presence of tunneling the smallest $k(E)$ is very small but nevertheless finite, the above inequality cannot be satisfied at any finite pressure, and the second-order limit becomes unreachable.

4. Decomposition of Formaldehyde into H + HCO

This will be referred to as the radical channel. From thermochemistry,²¹ mass spectrometry,²² and photochemical studies²³⁻²⁵ the energy barrier for this channel is deduced to be in the range 85–88 kcal/mol. Relative to the molecular-channel barrier, the radical-channel barrier appears to be²⁶ 1–4 kcal/mol higher. Ab initio calculations^{27,28} show no exit barrier for the radical channel, as one would expect for a fragmentation into two radicals, and therefore no tunneling is involved. Thus, the corresponding specific-energy rate constant $k(E)$ may be obtained by a routine RRKM calculation. The radical-channel barrier is indicated schematically in Figure 3.

The radical channel corresponds to type 1 reaction for which a loose transition state is indicated.¹⁰ In order to bracket the calculations, we shall take the loosest possible transition state, assuming free rotation of the HCO moiety in H $\cdots$ HCO, with vibrational frequencies and rotational constant B as given in Table I. The vibrational frequencies were chosen so as to yield zero-point energy of 2700 cm⁻¹ for HCO, as calculated by Goddard and Schaefer.²⁷ The rotational constant is taken to be slightly smaller than that of formaldehyde.⁴

5. Coupled-Channel Decomposition of Formaldehyde into CO + H₂ and H + HCO

The calculations of the thermal rate constant k_{uni} and its activation energy E_a presented so far refer to the hypothetical case when formaldehyde is assumed to decompose exclusively via the molecular channel. In reality, when energy is sufficiently large, both the molecular and radical channels are open and interact, which must be taken into account when calculating k_{uni} and E_a . This can

be accomplished by modifying eq 19 as follows:²⁹

$$k_{\text{uni}(i)} = \frac{\omega}{Q} \int_{E_{0i}}^{\infty} \frac{k_i(E)}{K(E) + \omega} N(E) e^{-E/kT} dE \quad (23)$$

where $k_{\text{uni}(i)}$ is k_{uni} for channel i , $k_i(E)$ is its specific-energy rate constant, E_{0i} is the energy barrier for channel i , and $K(E) = \sum_i k_i(E)$ is the total rate constant for decomposition into all channels at energy E . The activation energy corresponding to $k_{\text{uni}(i)}$ then follows by taking the logarithmic derivative of $k_{\text{uni}(i)}$ as indicated in eq 20. Note that channel coupling has no effect on k_{∞} (and hence also on $E_{a\infty}$), which is still given by eq 4, to which eq 23 reduces for $K(E) \ll \omega$.

Experimental measurements in a thermal system therefore yield the experimental counterparts of $k_{\text{uni}(i)}$. Regarding the radical-channel rate, there appears to be good agreement between two sets of shock-tube data: in the vicinity of 2000 K and at total concentrations of $\sim 10^{-5}$ mol/cm³ in argon, the reported rate constants for the reaction

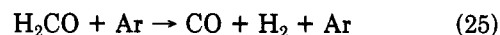


are (in cm³/(mol s))

Dean et al.³⁰: $k_{\text{rad}} = 3.3 \times 10^{16} \exp(-81.0 \text{ kcal}/RT)$

Just³¹: $k_{\text{rad}} = 1.7 \times 10^{16} \exp(-80.9 \text{ kcal}/RT)$

For the molecular channel, the experimental data are less clear-cut: Dean et al.³² report failure to detect the presence of molecular-channel decomposition in their experiments, but Just has pointed out³³ that their data could equally well be interpreted on the basis of a two-channel (molecular + radical) mechanism. Later Just³¹ reported for the reaction



under the same experimental conditions as for reaction 24, in cm³/(mol s)

$$k_{\text{molec}} \sim 1.4 \times 10^{16} \exp(-71.5 \text{ kcal}/RT)$$

with the additional information that, whereas (at 2000 K, $\sim 10^{-5}$ mol/cm³ total concentration) reaction 24 appears to be at its low-pressure limit, reaction 25 is still in the falloff regime.

In trying to reproduce these results by calculation, we shall assume that the collision efficiency (λ) of argon at 2000 K is $\lambda = 0.1$, i.e., that the effective collision cross section for CH₂O–Ar collisions is 1/10 of the strong-collision cross section. This is a crude way of accounting for weak-collision effects, which is justified in the present case by several other and more important uncertainties, as will become clear shortly. Preliminary calculations revealed very quickly that it is impossible to reproduce the experimental value of k_{molec} with the original theoretical threshold of $E_0 = 87$ kcal/mol, the calculated rate being too small (cf. Figure 4); hence, a lower molecular-channel threshold is clearly indicated. After a few trials, the choice $E_0 = 81$ kcal/mol, $D = 94.38$ for the molecular channel and $E_0 = 86$ kcal/mol for the radical channel, and using the

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(28) D. M. Hayes and K. Morokuma, *Chem. Phys. Lett.*, **12**, 539 (1972).

(29) Th. Just and J. Troe, *J. Phys. Chem.*, **84**, 3068 (1980).

(30) A. M. Dean, R. L. Johnson, and D. C. Steiner, *Combust. Flame*, **37**, 41 (1980).

(31) Th. Just in "Shock Waves in Chemistry", A. Lifshitz, Ed., Marcel Dekker, New York, 1981, p 303.

(32) A. M. Dean, B. L. Craig, R. L. Johnson, M. C. Schultz, and E. E. Wang, *Sympt. (Int.) Combust.*, [Proc.], **17**, 577 (1979).

(33) Th. Just, Discussion comments in *Symp. (Int.) Combust.*, [Proc.], **17**, 584 (1979).

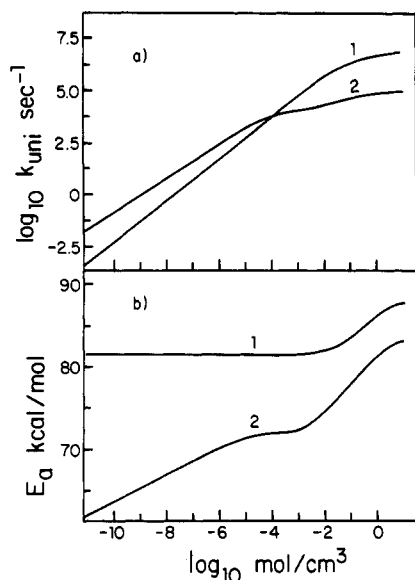


Figure 5. Falloff at 2000 K of thermal unimolecular rate constant k_{uni} (panel a) and activation energy E_a (panel b) for formaldehyde coupled-channel decomposition, calculated from eq 23 and 20, respectively, using $\lambda = 0.1$. Curve 1: radical channel, $E_0 = 86$ kcal/mol and transition-state frequencies from Table I. Curve 2: molecular channel, same parameters as in Figure 1.

frequencies given in Table I, yielded at 2000 K and 10^{-5} mol/cm³ ($\lambda = 0.1$) the following calculated results:

$$k_{\text{rad}} = 3.4 \times 10^{16} \exp(-81.4 \text{ kcal}/RT) \text{ cm}^3/(\text{mol s}) \quad (26)$$

$$k_{\text{molec}} = 7.8 \times 10^{15} \exp(-71.2 \text{ kcal}/RT) \text{ cm}^3/(\text{mol s}) \quad (27)$$

The full falloff plots for the rate constants and the activation energies are shown in Figure 5. These results were deemed close enough to experiment to make unproductive any attempt at a closer match. Although possible, a closer match would not increase the significance of the results in view of the uncertainties discussed below.

6. Discussion

The calculated results (eq 26 and 27) not only agree reasonably well with experiment with respect to the magnitude of the rate constants and activation energies but also show clearly that at the indicated concentrations the molecular channel is still in the falloff regime, while the radical channel is well at its low-pressure limit. Thus, the radical-channel result (eq 26) is essentially independent of the assumed value of λ , but this is not quite true of the molecular channel. If we assume $\lambda = 0.03$ (a plausible value), the result would be

$$k_{\text{molec}} = 6 \times 10^{15} \exp(-71.5 \text{ kcal}/RT) \text{ cm}^3/(\text{mol s}) \quad (28)$$

which is not too different from eq 27.

Although, as has been pointed out, the transition state for the radical channel is an extremely loose one, it requires a threshold near the low end of the range of experimental values in order to reproduce the experimental k_{rad} . Consequently any transition state tighter than the one used here would require a still lower radical threshold for the calculated k_{rad} to fit experiment. It is thus possible that the experimental k_{rad} , or the experimental value for the radical threshold, or both, are slightly too large.

In the case of the calculations for the molecular channel (eq 27 and 28), similar (or better) results could have been obtained with a different choice of E_0 , D , and transition-state frequencies. Of the three, transition-state frequencies have (within limits) only a small influence on the results,

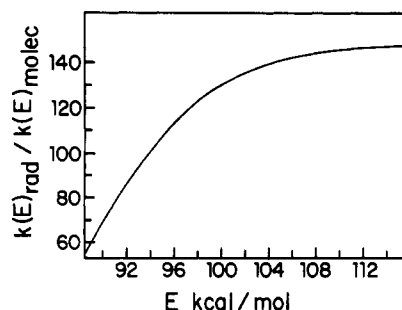


Figure 6. Calculated energy dependence of the ratio $k(E)(\text{radical channel})/k(E)(\text{molecular channel})$ in formaldehyde. Same parameters as in Figure 5.

and the other two work in opposite directions: increasing D and decreasing E_0 simultaneously leaves the result roughly unchanged. Thus, the choice $D \sim 80$, $E_0 \sim 83$ kcal/mol would yield k_{molec} not too different from eq 27. In order to get a better handle on D , it would be necessary to know from experiment the falloff of activation energy over a range of pressures (smaller D means faster falloff, cf. Figure 4), or to know k_{∞} where barrier permeability enters as a factor (cf. eq 9).

If the molecular-channel decomposition did not involve tunneling, pressure falloff of activation energy would be similar to that in the radical channel (cf. Figures 4b and 5b): (1) the reaction would be at its low-pressure limit at all concentrations below $\sim 10^{-5}$ mol/cm³ (assuming $\lambda = 0.1$); (2) the activation energy at the low-pressure limit would be 4–5 kcal/mol below threshold, i.e., not less than 77 kcal/mol if we assume that the threshold is at least 82 kcal/mol, which appears to be the consensus at present (see Conclusions). The cited experiments and the present calculations show $E_a \sim 71.5$ kcal/mol at $\sim 10^{-5}$ mol/cm³, which, if no tunneling takes place, would require (a) barrier no higher than ~ 77 kcal/mol and (b) no pressure dependence of E_a at 10^{-5} mol/cm³ and below. Thus, the cited experimental value for the molecular-channel activation energy and its pressure dependence at 10^{-5} mol/cm³ cannot be accounted for without invoking tunneling.

On the microscopic level, the ratio of the calculated specific-energy rate constant $k(E)$ for the radical and molecular channels (on which the results of eq 26 and 27 are based) is shown in Figure 6. Under collisionless conditions, Lee, Moore, et al.¹⁵ have determined $k(E)_{\text{rad}}/k(E)_{\text{molec}} \geq 10$ at $E = 100$ kcal/mol. The calculated result for this ratio is seen to be ~ 130 at the same energy, which is not incompatible with experiment, but does appear rather high. This, in turn, would indicate that the channel transition state is too loose, but, if it is, the experimental k_{rad} is a bit too high for a radical threshold in the 86–87 kcal/mol range.

The fact that the calculated radical $k(E)$ is almost 2 orders of magnitude larger than the molecular $k(E)$ over most of the energy range accounts for the dip in the activation energy falloff (Figure 5b), since, as is evident from eq 23, $K(E)$ suddenly increases by 2 orders of magnitude as soon as the radical channel opens. As usual, this is much less evident in the rate constant falloff.

Finally, the calculations yield a difference of 5 kcal/mol between the radical and molecular thresholds. This is slightly outside the range deduced from photochemical experiments, but could easily be adjusted downward by 1 or 2 kcal/mol if a slightly lower D and higher E_0 for the molecular channel and/or a lower E_0 and tighter transition state for the radical channel had been used, with virtually no influence on the computed k_{rad} and k_{molec} at $\sim 10^{-5}$ mol/cm³.

7. Conclusions

The present calculations show (but do not prove) that it is possible, in a general way, to reconcile photochemical and thermal data on the decomposition of formaldehyde by invoking a loose transition-state model for the radical channel and tunneling in the molecular channel. In fact, the low-pressure, molecular-channel thermal data cannot be interpreted in any other way if the barrier to decomposition exceeds ~ 77 kcal/mol.

Apart from this, there are few other definitive statements that can be made because, as it turns out, the original *ab initio* calculation⁵ produced a barrier for the molecular channel that is clearly too high, as these calculations usually do. Consequently $h\nu^*$ as well as the remaining molecular-channel transition-state frequencies and D are also in doubt. The problem thus appears undetermined. The combination $E_0 = 81$ kcal/mol, $D = 94.38$ used in obtaining the results in eq 27 implies $h\nu^* = 1886i$ for the imaginary frequency in the transition state. Since formaldehyde hydrogens are primarily involved in the tunneling, it is to be expected that this frequency would be high. Inasmuch as $D|h\nu^*| = 2\pi E_0$, then for thresholds in the vicinity of 80 kcal/mol the range of plausible values of D is not large if $|h\nu^*|$ is to be somewhere in the 2000–2500-cm⁻¹ range. Thus, while it would appear that the molecular-channel problem is undetermined, the possible range of adjustment on D and E_0 (the most important parameters) is narrow. The significance of the calculations presented here resides in the demonstration that it is possible to interpret the thermal data by using perhaps inexact but plausible values for the various parameters. In this context, the simple treatment of tunneling and total neglect of rotational effects is unlikely to cause errors sufficient to modify the general thrust of the conclusions. Among these, the most important is worth repeating: the one feature that distinguishes an "ordinary" thermal reaction from one involving tunneling is that both k_{uni} and E_a never reach their low-pressure limit; i.e., E_a in particular keeps falling off appreciably down to the lowest pressures. This feature is independent of any particular value of any of the parameters (except, perhaps, when D is large) but is also the most difficult to demonstrate experimentally.

It is also important to note that the much lower threshold for the molecular channel came out of the necessity to fit experiment and not as a result of an assumption. An *ab initio* calculation exists³⁴ in fact that puts the threshold at 79.6 kcal/mol. There appears to be a consensus at present³⁵ that the molecular-channel threshold is probably somewhere between 82 and 84 kcal/mol, a range easily accommodated in the present calculations by a minor change of other parameters.

The thermal data themselves are not free of doubt. The experimental value of k_{molec} used here is at present the only determination.³⁶ While there are the two cited shock-tube determinations of k_{rad} which both agree, Batt et al.³⁷ report from reinterpretation of bulb experiments at 800 K and $\sim 10^{-5}$ mol/cm³ $k_{rad} \sim 1.58 \times 10^{15} \exp(-77 \text{ kcal}/RT)$ cm³/(mol s), which is reminiscent of an early 2000 K shock-tube result of Schecker and Jost,³⁸ $k_{rad} \sim 5 \times 10^{16} \exp(-72 \text{ kcal}/RT)$ in the same units. According to the present calculations (cf. Figure 5b), such low activation energy could never be obtained for the radical channel, but might be that of the molecular channel, unless the presently accepted radical-channel barrier (i.e., the H-HCO bond energy) should turn out to be too high by ~ 10 kcal/mol or more.

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